Yuhai Wang

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Research Summary

Ph.D. student in **robot learning and visual representation learning**, building self-supervised models that let robots perceive, predict, and plan from pixels. Current focus: **latent world models** (JEPA) with structured, composable action representations for planning. Broader experience spans reinforcement learning for dexterous manipulation, 3D vision (NeRF), visual place recognition, and person re-identification, on simulation (Isaac Gym/Lab, MuJoCo, OGBench) and hardware (Franka, Allegro, ARX5, Unitree G1/Go2).

Education

Northeastern University — Boston, MA
Ph.D. in Civil and Environmental Engineering; Advisor: Gilbert Yang Ye — *Sep 2025 – expected 2029*

University of Southern California — Los Angeles, CA
M.S. in Analytics; Research with Daniel Seita — *Jan 2023 – Dec 2024*

Tiangong University — Tianjin, China
B.E. in Internet of Things Engineering; Research with Xuan Xiao — *Aug 2018 – May 2022*

Publications

Preprints and papers under review

1. **Yuhai Wang**, Jiawei Xia, Rongxuan Zhou, Xiao Hu, Yongliang Shi, Jing Du, Yang Ye. **PRISM: Prior-Guided Imagination Sampling in World Models.** *Preprint, 2026.* [pdf] [page] [code]
   Adopted as a baseline by SAGE (PKU), INTACT (ZJU) and RWM (Tsinghua IIIS), 2026.
2. **Yuhai Wang**, Haichao Zhang, Qi Yin, Rongxuan Zhou, Xiao Hu, Yangming Shi, Yang Ye. **EquiJEPA: Learning Action-Centered Latent Transition Geometry.** *Under review, 2026.* [page]
3. **Yuhai Wang**, Xiao Hu, Yangming Shi, Yang Ye. **DR-VPR: Dual-Branch Rotation-Robust Visual Place Recognition.** *Preprint, 2025.* [pdf] [code]

Peer-reviewed

4. **Yuhai Wang**, Chenghao Wang, Xianyao Li, Zening Teng, Xiao Hu, Rongxuan Zhou, Alireza Ramezani, Jing Du, Yang Ye. **Thinking on Its Feet: Terrain-Aware Hierarchical Control with Proprioceptive Feedback for Humanoid Navigation.** *IEEE-RAS Humanoids 2026.* [page]
5. Xianyao Li, **Yuhai Wang**, Xiao Hu, Kaleb Smith, Gilbert Yang Ye, Eric Jing Du. **When to Personalize Household Object Search: A Rigidity-Gated Hybrid Policy.** *IROS 2026.* [pdf]
6. **Yuhai Wang**, Maryam Pishgar. **Dynamic Token Selective Transformer for Aerial-Ground Person Re-Identification.** *ICME 2025.* [pdf] [page] [code]
7. Hao Jiang, **Yuhai Wang***, Hanyang Zhou*, Daniel Seita. **Learning to Singulate Objects in Packed Environments Using a Dexterous Hand.** *ISRR 2024.* [pdf] [page] [code]
8. Baijun Ye, Caiyun Liu, Xiaoyu Ye, Yuantao Chen, **Yuhai Wang**, Zike Yan, Yongliang Shi, Hao Zhao, Guyue Zhou. **Blending Distributed NeRFs with Tri-stage Robust Pose Optimization.** *IROS 2024.* [pdf]
9. Yongjiang Xue, Xichen Yuan, **Yuhai Wang**, Yang Yang, Siyu Lu, Bo Zhang, Juezhu Lai, Jianming Wang, Xuan Xiao. **Lywal: A Leg-Wheel Transformable Quadruped Robot with Picking Up and Transport Functions.** *ICRA 2021.* [pdf] [video]

* equal contribution

Research Experience

CRAFT Lab, Northeastern University — Boston, MA
Graduate Research Assistant, advised by Prof. Gilbert Yang Ye — *Sep 2025 – Present*

- **EquiJEPA:** Designed a state-blind action-effect objective for JEPA world models that makes the same action produce a consistent latent transition across states, turning actions into reusable, composable latent prototypes. Evaluated on four simulated tasks: 3.5× higher success on unseen action combinations (11.5% → 40.1%, two-dot world), 34% lower error when reusing prototypes at new states (Push-T), and label-free wall detection from interaction residuals (Maze, AUC 0.66 → 0.94).

- **PRISM:** Attached a lightweight action-prior head to a frozen JEPA world-model encoder and fused its state-conditioned Gaussian into the MPPI sampling distribution with a closed-form, precision-weighted update. Improved success by 35 points on Cube and 32 points on Push-T over vanilla world-model MPC without significant inference overhead; code and project page released.
- **TOIF:** Built a three-tier humanoid navigation system (LLM planner, VLA navigator, RL locomotion) that feeds proprioceptive terrain anomalies back to the planner as language for rerouting. Raised simulated success over the strongest baseline (45% → 65%, 35% → 60%); deployed on a physical Unitree G1 (real-world detector AUC 0.81–0.94).
- Designed DR-VPR, a dual-branch visual place recognition method that stays robust under large rotation and viewpoint change.
- Co-developed a rigidity-gated hybrid policy that decides when to personalize household object search (IROS 2026).

SLURM Lab, University of Southern California — Los Angeles, CA
Research Assistant, advised by Prof. Daniel Seita — *Sep 2023 – May 2025*

- Built an Isaac Gym RL framework for dexterous object singulation in clutter (displacement-based state, multi-phase training); validated on Allegro + Franka hardware (ISRR 2024).
- Designed a top-k token-selective Transformer for aerial-ground person re-identification (ICME 2025).

Institute for AI Industry Research (AIR), Tsinghua University — Beijing, China / Remote
Research Engineer, later Research Assistant (remote); advised by Guyue Zhou — *Aug 2021 – Feb 2022; Apr – Sep 2023*

- Built a distributed NeRF system with three-stage robust pose optimization and fused NeRFs across coordinate frames (IROS 2024).
- Controlled the ARX5 arm (ROS/MoveIt, ArUco servoing); built ICRA 2022 RoboMaster Sim2Real env.

Robotics Research Lab, Tiangong University — Tianjin, China
Research Assistant, advised by Prof. Xuan Xiao — *Oct 2019 – Aug 2021*

- Designed a leg-wheel quadruped: C kinematics, Webots gait simulation, hardware tests (ICRA 2021).

Skills

Languages	Python, C/C++, MATLAB
ML	PyTorch; self-supervised representation learning (JEPA), reinforcement learning, sparse autoencoders
Robotics	ROS / MoveIt, Isaac Gym / Lab, MuJoCo, OGBench, Webots; sampling-based planning (MPPI / CEM)
Hardware	Franka, Allegro Hand, ARX5, Unitree G1 / Go2, Intel RealSense

Service & Teaching

Reviewer	CoRL 2026, IROS 2026, ICME 2025, IEEE THMS (2025, 2026), ASCE JCEM 2026, ICRA'24 Agile Robotics WS
Competitions	WBCD Competition @ ICRA 2025, hardware sponsor
Teaching & Talks	Graduate TA, ISE 534, USC (Spring 2024); CEE Graduate Research Expo, Northeastern (2025)

Honors & Awards

CEE Department Fellowship, Northeastern (2025, 2026)
First Prize, National Challenge Cup (2021)
Honorable Mention, MCM (2020)
President's Scholarship, Tiangong University (2019, 2020)